MILITARY
SPECIFICATION

MIL-T-12242A (EL)
20 June 1963
Superseding
MIL-T-12242(SigC)
6 October 1952

TEST SET, TELEPHONE TS-716( )/U

1. SCOPE

1.1 This specification covers Test Set, Telephone TS-716( )/U, a portable, transistorized multi-purpose instrument designed to evaluate the electrical and accoustical performance of all types of transducers and telephone communication equipment. (See 6.3).

2. APPLICABLE DOCUMENT

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

QQ-S-781	Strapping, Flat; Steel
UU-T-111	Tape; Paper, Gummed (Sealing and Securing)
PPP-B-601	Boxes, Wood, Cleated plywood.
PPP-B-621	Boxes, Wood, Nailed and Lock-Corner
PPP-B-636	Boxes, Fiber, Solid.
PPP-F-320	Fiberboard Sheet, Stock and Cut Shapes
PPP-P-291	Paperboard, Wrapping, Cushioning
PPP-T-76	Tape, Pressure-sensitive Adhesive, Paper, Water resistant.

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MIL-P-116	Preservation, Methods of
MIL-B-117	Bags, Interior Packaging
MIL-P-11268	Parts, Materials, and Processes Used in Electronic Communication Equipment.

FSC 6625

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MIL-I-11748	Interference Reduction for Electrical Equipment.
MIL-M-13231	Marking of Electronic Items
MIL-F-14072	Finishes for Ground Signal Equipment

STANDARD

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MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	Marking for Shipment and Storage.
MIL-STD-169	Extreme-Temperature Cycle.
MIL-STD-170	Moisture Resistance Test Cycle for Ground Signal Equipment
MIL-STD-252	Wired Equipment, Classification of Visual and Mechanical Defects

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer. Both the title and number or symbol should be stipulated when requesting copies.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

AMERICAN STANDARDS ASSOCIATION

ASA Z24.4-1949	Pressure Calibration of Laboratory Standard Pressure Microphone

(Application for copies should be addressed to the American Standards Association, Inc., 10 East 40th St., New York 16, N.Y.)

3. REQUIREMENTS

3.1 Description. - Test Set, Telephone TS-716( )/U is a portable, self-calibrating, multi-purpose instrument used to test and evaluate on a "GOOD" or "BAD" type basis the electrical and acoustical performance of telephone equipment. It performs the following functions:

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3.1 Description.(cont'd)

(a) Transmit and receive efficiency tests of complete telephone sets.

(b) Speed and percent break of telephone dials.

(c) Test of line level and click suppression varistors.

(d) Test of ringers.

(e) Test of telephone hand generators.

(f) Frequency response and sensitivity tests of earphones, carbon and dynamic microphones, sound-powered telephones and special purpose narrow band transducers.

(g) Test of insulation resistance.

(h) Continuity checks of circuit wiring.

3.2 Procurement model. - A procurement model of the equipment will be available for inspection by prospective bidders and will be lent to the contractor. Unless otherwise specified herein or in the invitation for bids, physical construction of the equipment shall conform to the model and the equipment shall incorporate all features of the model. (NOTE: In case of conflict between specified performance characteristics for the equipment and the performance of the model, the specified performance characteristics govern.)

3.3 Preproduction samples. - The contractor shall furnish 2 preproduction samples of Test Set, Telephone TS-713( )/U for approval, provided the invitation for bids and the contract require samples. (See 4.3 and 6.2).

3.4 Parts, materials and processes general. - In addition to the requirements of this specification, the requirements of Specification MIL-P-11268 including the selection requirements therein, shall apply.

3.5 Finish, protective. - The equipment shall be given protective finish in accordance with Specification MIL-F-14072. This includes finish of hardware, such as handles, hinges, screws, etc., and necessary touch-up after mounting. The final paint film on type I surfaces shall be final film E, semigloss, light-gray enamel conforming to Specification MIL-F-14072. (See 4.4).

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3.6 Marking.-

3.6.1 General.- Marking shall conform to Specification MIL-M-13231 (See 4.4). Front panel marking shall be group I as described in that specification.

3.6.2 Visibility.- Wherever practicable, parts shall be so mounted that their identification markings will be readily visible with minimum disassembly of the equipment.

3.6.3 Serial numbers.- Serial numbers are required.

3.7 Operating power.- The telephone test set shall comply with specification requirements when operated from an external power source generating 250 watts at line voltages of 115 volts ±15 percent over the frequency range of 50 to 60 cps. Power-on condition shall be indicated by the front panel power-on lamp.

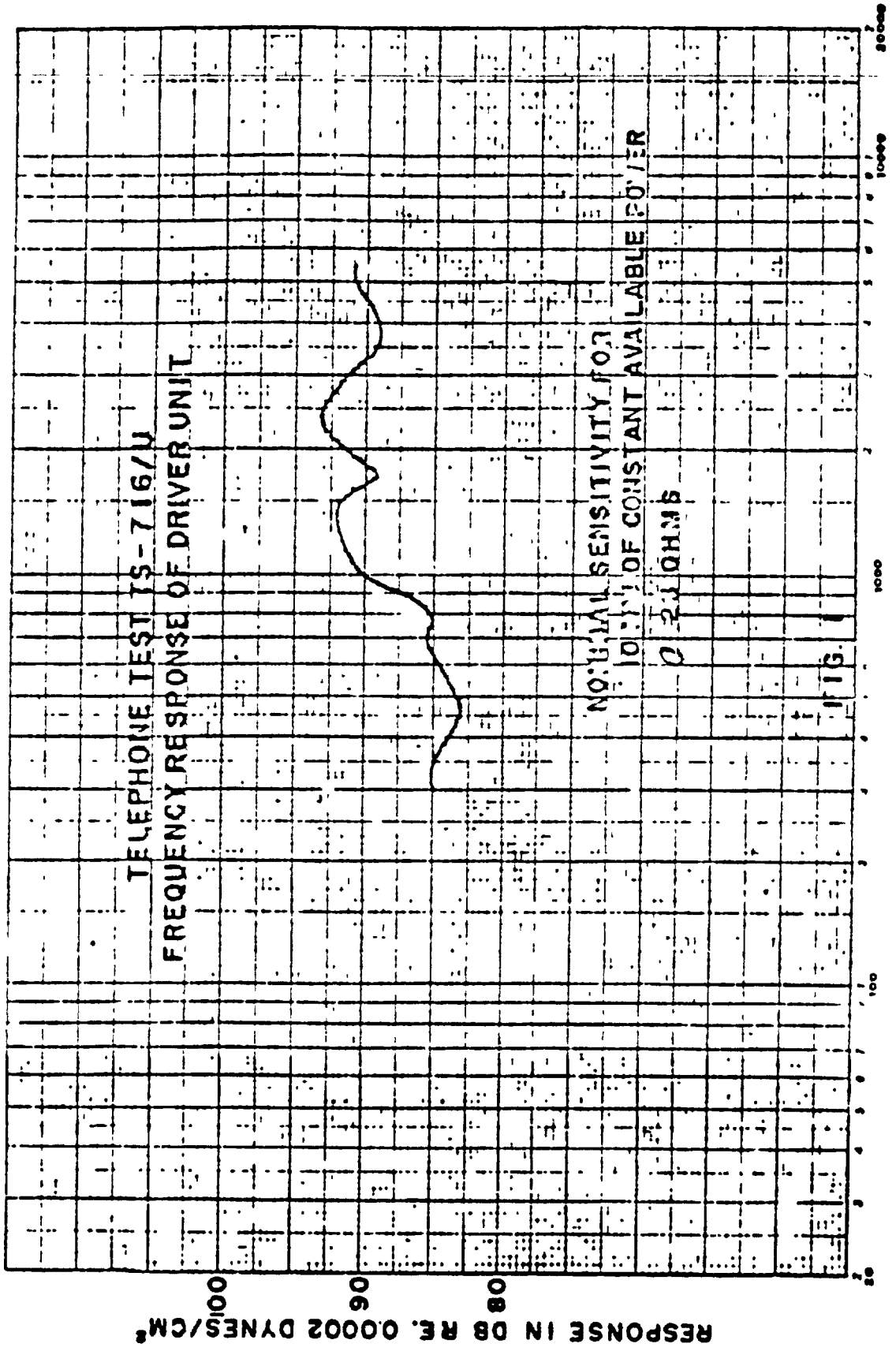
3.8 Driver unit.-

3.8.1 Driver unit response.- The output of the driver unit shall be not less than 90 db above a reference level of .0002 microbars when [illegible] milliwatts rms power is applied to the driver unit terminals. The response of the driver unit over the frequency range from 300 to 5100 cps shall not deviate more than ±3 db from the response curve shown in Figure 1.

3.8.2 Impedance.- The impedance of this driver unit at 1000 cps shall be 23 ohms ±10% as determined by the test of 4.8.2.

3.8.3 Distortion.- The acoustic output of the driver unit shall have not more than 5% total harmonic distortion over the audio frequency range of 300 to 5100 cps when measured as specified in 4.8.3.

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3.9 Coupler microphone. -

3.9.1 Coupler microphone response. - The frequency response of the microphone shall be essentially smooth from 300 to 5100 cps. The minimum power output of the microphone shall be -50 dbm at 1000 cps when measured in accordance with 4.9.1. In addition, the response shall not deviate more than ±3 db from the response curve shown in Figure 2. During this test the microphone shall be terminated with a non-inductive of 7 ohms.

3.9.2 Distortion. - Total harmonic distortion shall not exceed 5% at a sound pressure level of 103 db above .0002 microbars over the frequency range of 300 to 5100 cps.

3.10 Microphone pre-amplifier. -

3.10.1 Frequency response and gain. - The frequency response of this pre-amplifier shall be essentially flat from 300 cps to 5100 cps and shall provide a gain of 60 db ±2 db over that frequency range.

3.10.2 Distortion. - Total harmonic distortion at the rate output of the pre-amplifier shall not exceed 2% over the frequency range of 300 cps to 5100 cps.

3.11 Meter amplifier. -

3.11.1 Gain. - The meter amplifier shall provide a nominal gain of 65 db when measured as in 4.11.1.

3.11.2 Meter. - The yellow, red, and green color bands on face of panel meter shall indicate db values as follows:

Yellow band	±6.0 db
Red band	±3.0 db
Green band	±1.5 db

3.12 Sound source supply. - The sound source supply shall provide a sound pressure of not less than 20 microbars at the screen of the driver unit when measured as in 4.12.

3.13 Hand generator test circuit. - The hand generator test circuit shall test generators that deliver 50 volts to a load of 1500 ohms. The meter circuit shall be used to indicate when the generator is producing its rated load voltage.

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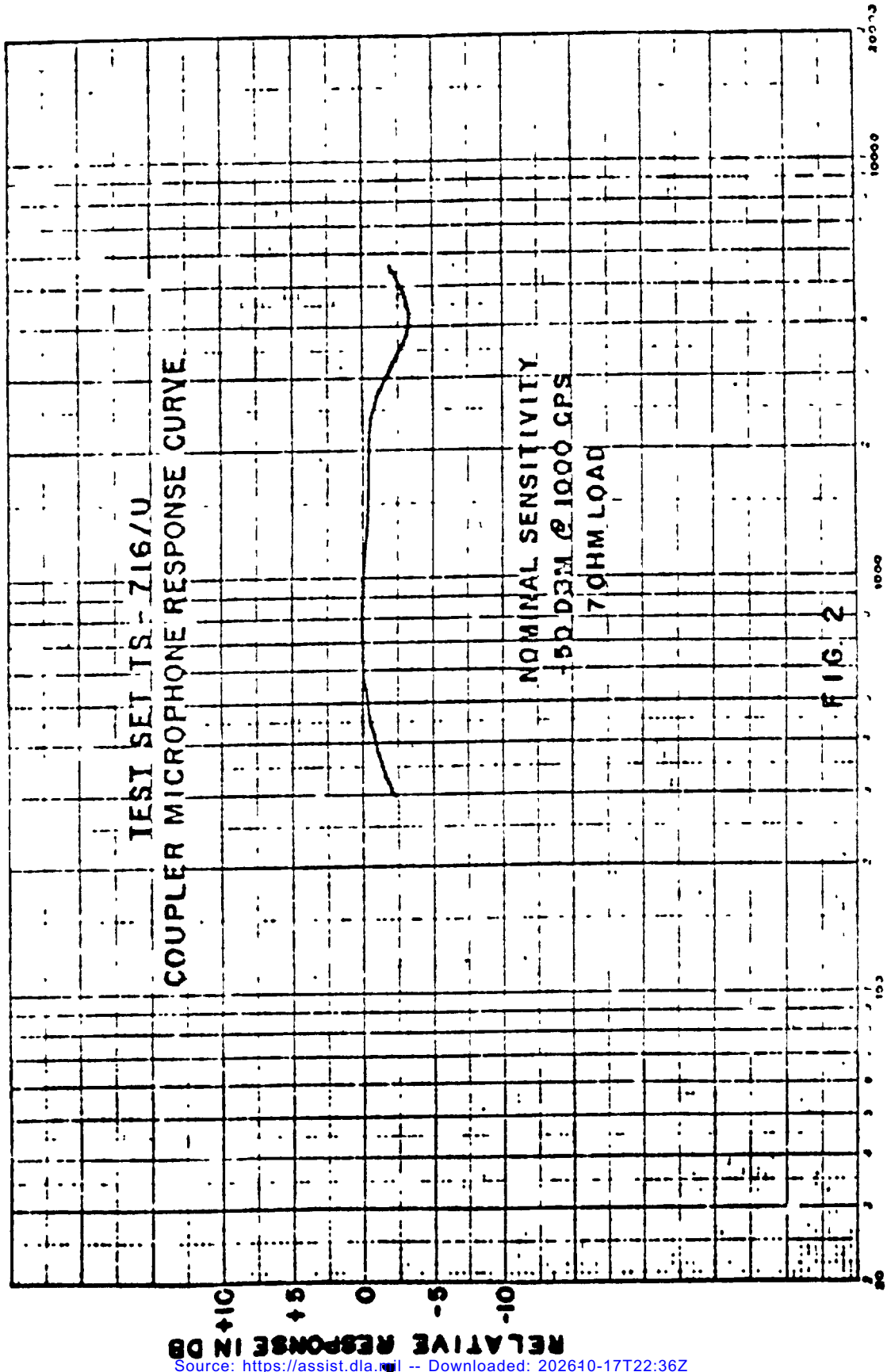


FIG. 2

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3.14 Ringer test circuit. - The ringer test circuit shall be capable of testing ringers by supplying them with their minimum 20 cps operating currents for ringers of 2400, 5300 and 12,000 ohms impedance.

3.15 Varistor test circuit. - The varistor test circuit shall provide an indication of the efficacy of varistors used in telephone equipment.

3.16 Earphone test circuit. - The earphone test circuit in combination with the coupler-microphone shall be capable of testing earphones with the following impedance; 15, 125, 300, 1000 and 8000 ohms.

3.17 Dynamic microphone test circuit. - The dynamic microphone test circuit in combination with the driver unit shall be capable of testing dynamic microphones with the following impedance values; 4,150, 1000 and 50,000 ohms at 1000 cps.

3.18 Filter circuit. - The filter circuit, with proper switching, shall connect band-pass filters between microphone pre-amplifier output and meter amplifier input for frequency response measurements of the various earphones and microphones. The filter passband frequencies shall be as follows:

1. Low 300 to 1100 cps
2. Center 1100 to 2200 cps
3. High 2200 to 5100 cps
4. Narrow Band 975 to 1025 cps

3.19 Carbon microphone test circuit. - The carbon microphone test circuit, in combination with the driver unit, shall test carbon microphones with the following impedance values, 20, 40, 100 and 150 ohms, and shall provide direct current values from 10 to 100 milliamperes.

3.20 Dial percent break test circuit. - This circuit shall indicate a percent break of 61.5±2 percent with an accuracy of ±0.5 percent.

3.21 Dial speed test circuit. - The dial speed test circuit shall be independent of the dial percent break test and shall indicate a dial speed of 10 ±2 pulses per second with an accuracy of ±1/2 pulse per second.

3.22 Insulation resistance test circuit. - The insulation resistance test circuit shall indicate only whether the insulation resistance is greater or less than 1 megohm.

3.23 Continuity test circuit. - The continuity test circuit shall provide an audible and visual signal when the continuity test terminals are connected across a circuit whose resistance is less than 10 ohms.

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3.24 Telephone Efficiency - Send circuitry.- Telephone efficiency - Send circuitry shall test the transmitting efficiency of a complete telephone set. (See 4.13.20).

3.25 Telephone Efficiency - Receive circuitry. - Telephone efficiency - Receive circuitry shall test the receiving efficiency of a complete telephone set. (See 4.13.21).

3.26 Service conditions. - The equipment shall meet the following service conditions--where a test is referenced, meeting the test shall be considered as compliance with the requirement:

3.26.1 Operation. - Continuous use for a period of 10 years, under the worldwide environmetal conditions specified in 3.26.2 through 3.26.6, with a duty cycle of 16 hours on and 8 hours off and with no more than normal maintenance and replacement of parts. Where the contractor is required to make a selection of parts, materials, processes, construction methods, etc, he shall be guided by this requirement. Approval of the preproduction sample (see 3.3) shall be considered as compliance with this requirement.

3.26.2 Temperature (see 4.14.1). -

(a) Operating: Ambient temperature in the range of +131°F to +32° F. (The 131° F temperature includes effect of sun-load.) Exposure at the high temperature extreme not to exceed 4 hours, and at the low temperature extreme not to exceed 72 hours, at any one time.

(b) Nonoperating: Exposure in the range of +155° F to -65° F; exposure at the high temperature extreme not to exceed 4 hours, and at the low temperature extreme not to exceed 24 hours, at any one time.

3.26.3 Relative humidity. - Exposure to 97 percent relative humidity for 20 hours and 100 percent relative humidity (with condensation) for 4 hours for each 24 hour period over the period of years stated in 3.26.1. Meeting the test of 4.14.2 shall be considered compliance with this requirement.

3.26.4 Elevation (see 4.14.3). -

(a) Operating. - Up to 10,000 feet above sea level.

(b) Nonoperating. - Up to 50,000 feet above sea level.

3.26.5 Vibration (see 4.14.4). - The amplitude of vibration of any part, subassembly or structural member of the equipment shall not exceed twice the amplitude of the vibration applied to the equipment at any frequency between 10 and 55 cycles per second.

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3.26.6 Shock; bench-handling.- The equipment shall meet specified performance after subjection to the test of 4.14.5.

3.27 Suppression of radio interference.- The equipment shall meet the emanation and susceptibility limits of Specification MIL-I-11748 for class 1c equipment. Application of suppression components and techniques shall conform to that specification. If the contractor considers it desirable to modify the equipment proper in order to facilitate conformance with the specified emanation and susceptibility limits, he shall obtain approval from the contracting officer before proceeding with the modification. (See 4.4)

3.28 Interchangeability.- Like units, assemblies, subassemblies, and replaceable parts shall be physically and functionally interchangeable, without modification of such items or of the equipment. Individual items shall not be hand-picked for fit or performance. Reliance shall not be placed on any unspecified dimension, rating, characteristics, etc. (See 4.15)

3.29 Preconditioning.- The equipment shall be capable of meeting the requirements herein, without subsequent processing, after subjection to the bounce preconditioning of 4.6.

3.30 Technical literature and running spare parts.- Technical literature and running spare parts shall be furnished if specified in the contract. Running spare parts shall be identical to corresponding parts in the equipment furnished on the order. (See 6.2)

3.31 Workmanship.- The Test Set, Telephone TS-716( )/U shall be manufactured and assembled in accordance with the applicable portions of the following paragraphs:

In Specification MIL-P-11268

General requirements for plastic material and plastic parts.
Wiring and cabling, including:
Process for soldering.
Cleaning of equipment.
Riveting.

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3.31 Workmanship.- (cont'd)

Stock.
Protection.
Clearance
Splicing.
Connections, general
Grounding, general.
Shielding on wire and cable.
Flux and cleaning agents for soldering.
General requirements for securing of parts.
Cleaning of Welding.
Process for welding.
Mechanical operation of controls.
Screws.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. - Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right, to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 Classification of inspection. - Inspection shall be classified as follows:

(a) Preproduction inspection (does not include preparation for delivery). (See 4.3)

(b) Inspection covered by subsidiary documents (See 4.4).

(c) Quality conformance inspection.

(1) Quality conformance inspection of equipment before preparation for delivery. (See 4.5).

(2) Quality conformance inspection of preparation for delivery.(See 4.17)

4.3 Preproduction inspection. - This inspection will be performed by the Government unless otherwise specified in the contract. It shall consist of the preproduction inspection specified in Table I, the inspection specified in the subsidiary documents covering the items listed in 4.4, and the inspection specified for group A, group B and group C (See Tables II, III and IV respectively).

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4.3 Preproduction inspection. - (cont'd)

Table I Preproduction Inspection

Inspection	Requ. Para.	Insp. Para.
Temperature	3.26.2	4.14.1
Moisture resistance	3.26.3	4.14.2
Elevation	3.26.4	4.14.3
Vibration	3.26.5	4.14.4

4.4 Inspection covered by subsidiary documents. The following shall be inspected under the applicable subsidiary documents as part of the acceptance before preparation for delivery:

Item	Where Required
Finish	3.5
Marking	3.6
Suppression	3.27

4.5 Quality conformance inspection of equipment before preparation for delivery. - The contractor shall perform the inspection specified in 4.4 and 4.5.1 through 4.5.3.2. This does not relieve the contractor of his responsibility for performing any additional inspection which is necessary to control the quality of the product and to assure compliance with all specification requirements. The Government will review and evaluate, the Contractor's inspection procedures and examine the contractor's inspection records. In addition the Government--at its discretion--may perform all or any part of the specified inspection, to verify the contractor's compliance with specified requirements.(See 6.4). Test equipment for Government verification inspection shall be made available by the contractor. Each unit, which will be subjected to Group A or group B inspection,except preproduction samples, shall be preconditioned after final assembly.(See 3.27).

4.5.1 Group A inspection. - This inspection, including sampling, shall conform to table II and the ordinary inspection procedures of Standard MIL-STD-105. Group A inspection shall be performed in any order which is satisfactory to the Government.

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Table II - Group A Inspection

Inspection	Req Para	Insp Para	AQL Major	AQL Minor
Visual & Mechanical	3.31	4.16	1.5 DPHU	6.5 DPHU **
Electrical				
Sound Source Supply	3.12	4.12, 4.13.2	1.0%	*
Circuit Tests (Continuity)	3.15 thru 3.25	4.13 thru 4.13.22	1.5 DPHU	*
Operating Power (Nominal)	3.7	4.7.1	combined for the group	
Driver Unit				
Response	3.8.1	4.8.1		
Impedance	3.8.2	4.8.2		
Distortion	3.8.3	4.8.3	1.5%	
Coupler Microphone				*
Response	3.9.1	4.9.1		
Distortion	3.9.2	4.9.2		
Microphone Preamplifier				
Frequency Response & Gain	3.10.1	4.10.1		
Distortion	3.10.2	4.10.2		
Meter Amplifier				
Gain	3.11.1	4.11.1		
Meter (color band)	3.11.2	4.11.2		

* All electrical defects are considered major.

** Defects per hundred units.

4.5.2 Group B Inspection.- This inspection, including sampling, shall conform to Table III and to the special procedures for small-sample inspection of Standard MIL-STD-105. The AQL shall be 6.5 percent defective and the inspection level shall be L-7 for normal and tightened inspection and L-5 for reduced inspection. The reduced inspection procedure shall be R-1. Group B inspection shall normally be performed on inspection lots that have passed group A inspection and on samples selected from units that have been subjected to and met the group A inspection.

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4.5.2.1 Order of inspection within group B.- Group B inspection shall be performed in any order which is satisfactory to the Government.

Table III - Group B inspection

Inspection	Req Para.	Test Para	AQL
Operating Power (Varying)	3.7	4.7.2	
Driver Unit			
Response	3.8.1	4.8.1	
Impedance	3.8.2	4.8.2	
Distortion	3.8.3	4.8.3	
Coupler Microphone			
Response	3.9.1	4.9.1	6.5% DPHU
Distortion	3.9.2	4.9.2	
Microphone Preamplifier			
Frequency Response & Gain	3.10.1	4.10.1	
Distortion	3.10.2	4.10.2	
Meter Amplifier			
Gain	3.11.1	4.11.1	
Meter (Color Band)	3.11.2	4.11.2	
Interchangeability	3.28	4.15	6.5%

4.5.3 Group C inspection.- This inspection shall be as listed in Table IV, and shall normally be performed on sample units that have been subjected to and met group A and group B inspection.

Table IV - Group C inspection

Inspection	Req Para	Test Para
Shock, Bench Handling	3.26.6	4.14.5

4.5.3.1 Sampling for inspection of equipment.- Three tests sets for group C inspection shall be selected from each 100 units or fraction thereof produced without regard to their quality, except that the units inspected at the start of the contract shall be selected from the first units produced.

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4.5.3.2 Noncompliance.- If a sample unit fails group C inspection, the contractor shall immediately investigate the cause of failure and shall report to the Government Inspector the results thereof and details of the corrective action taken on the process and all units of product which were manufactured with the same conditions, materials, processes, etc. If the Government inspector does not consider that the corrective action will enable the product to meet specified requirements, or if the contractor cannot determine the cause of failure, the matter shall be referred to the Contracting Officer. (See 6.5)

4.6 Bounce preconditioning.- The unit shall be placed in its normal operating position on the table of the Package Tester, Type 1000-SC, as made by the L.A.B. Corporation, Skaneateles, New York, or equal. The package tester, shafts in phase, shall have a speed such that it is just possible to insert a 1/32-inch-thick strip of material under one corner or edge of the unit to a distance of 3 inches as the unit bounces. The unit shall be subjected to this preconditioning for 1 minute. After bounce preconditioning the unit shall not be repaired, alined, cleaned, or otherwise changed prior to quality conformance inspection. (see 3.29)

4.7 Operating Power.- The equipment shall be connected to an external power source of 115 volts a.c., that can be varied ±15% and the frequency can be varied 50 to 60 cps. The tests of 4.8, 4.9, 4.10 and 4.11 shall be performed at the following voltages and frequencies:

4.7.1 Nominal operating power.- 115 volts a.c., 60 c.p.s.

4.7.2 Varying operating power.- 100 volts a.c. and 130 volts a.c., 50 c.p.s. and 60 c.p.s.

4.8 Driver unit tests.-

4.8.1 Response test.- Available power frequency response measurements shall be made starting at 300 cps and extending through 5100 cps, in sufficient detail to establish definitely the shape of the curve. A Western Electric Company 640AA Condenser Microphone or its equivalent shall be used for measuring sound pressure.

4.8.1.1 Test equipment.- The test equipment used for the response test shall meet the following requirements.

(a) Calibrating microphone.- A Western Electric Co. 640AA condenser microphone, or equal shall be used for measuring sound pressure. It shall be calibrated by the reciprocity method in accordance with A.S.A Standard Z24.4 "Pressure Calibration of Laboratory Standard Pressure Microphones".

(b) Audio oscillator.- The audio oscillator shall have a frequency range of at least 100 to 10,000 cps, shall have a high degree of stability in both output voltage and frequency, and shall have a waveform distortion of less than 2 percent.

(c) Vacuum-tube voltmeter (VTVM).- The vacuum-tube voltmeters used must have flat frequency response (±1 db) from at least 100 to 10,000 cps and must be capable of measuring voltage from 0.001 rms to 100 volts or more.

(d) Microphone pre-amplifier.- The microphone pre-amplifier shall have a flat response (±1 db) over a frequency range of at least 100 to 10,000 cps, shall have a high degree of stability, and shall have distortion of less than 2 percent.

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4.8.1.2 Testing procedure. - The driver unit under test shall be connected to the test circuit as shown in Figure 3. The output from the oscillator at each test frequency shall be adjusted to 0.96 volts rms as measured by vacuum tube voltmeter (VTVM) #1. The output from this calibrated microphone and its associated pre-amplifier unit shall be measured as indicated by the reading of VTVM #2. This reading shall be converted to the equivalent db value above .0002 microbars using the most recent available calibration curve for the test microphone in use. The operations set forth above shall be performed from 300 cps to 1000 cps inclusive, with measurements made at each 100 cps increment, and from 1000 cps to 5100 cps inclusive, with measurements at each 250 cps increment. The response of the driver unit shall meet the requirements specified in 3.8.1.

4.8.2 Impedance test. - The driver unit shall be connected in series with a variable resistor, or a suitably calibrated resistor across the output of an audio oscillator set at 1000 cps. The resistance shall be adjusted until the voltage drops across the resistor and the driver unit are equal and the adjusted value of the resistance shall be determined. The drive unit impedance shall be considered as numerically equal to the adjusted value of the variable resistor and shall comply with 3.8.2.

4.8.3 Distortion test. - The driver unit shall be connected as described in 4.8.1.2. The output of the oscillator shall be adjusted to 0.96 rms volts at 300 cps across the series combination of the 23 ohm resistor connected to the driver unit. The harmonic distortion shall be measured with a Hewlett-Packard Model 330-B Total Noise and Distortion Meter, or equal, connected across the output of the calibrated microphone and preamplifier as in Figure 3. The distortion measurement shall be repeated at 400 cps and at sufficient points between 400 cps and 5100 cps to determine the frequency where maximum distortion exists. Total harmonic distortion shall meet the requirements of 3.8.3.

4.9 Microphone tests. -

4.9.1 Response test. - The response of the microphone shall be measured by means of a suitable a-c electronic voltmeter, having an input impedance of at least 1/2 megohm. The response of the microphone shall be determined using a constant sound pressure of 28 microbars at the point where the microphone is to be placed. The driver unit used to obtain the desired sound pressure input, (Western Electric Co. Type No. 555W, or equal) shall have been previously calibrated by means of a condenser microphone such as Western Electric Co. 640AA, or equal, (calibrated by "reciprocity method" in accordance with A.S.A. Standard Z24.4-1949 "Pressure Calibration of Laboratory Standard Pressure Microphones".) The sample dynamic microphone shall be positioned with its diaphragm 1/2 inch directly in front of the driver unit with the diaphragm of the microphone parallel to the driver unit. The voltage - frequency response of the microphone shall be measured across a non-inductive load resistance of 7 ohms. The response of the microphone shall be tested in the frequency range of 300 to 5100 cycles per second at approximately the following

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4.9.1 Response test.- (cont'd) increments: 100 cps increments from 300 to 1000 cps and 250 cps increments from 1000 to 5100 cps. The microphone shall meet the requirements of 3.9.1.

4.9.2 Distortion. - Harmonic distortion shall be determined using the same equipment and circuitry as used in 4.9.1 except that the constant sound pressure shall be 103 db referenced at .0002 microbars and the output of the microphone terminated in 7 ohms shall be connected to a Hewlett-Packard Distortion Analyzer Model 330C or equal for distortion measurements. The distortion shall not be greater than 5%. Measurements will be made from 300 cps to 1000 cps inclusive at each 100 cps increment and from 1000 to 5100 cps inclusive at each 500 cps increment.

4.10 Microphone pre-amplifier tests. -

4.10.1 Frequency response and gain. - With equipment turned on, connect the 600 ohm output of the audio oscillator to the front panel terminals marked "Microphone-Other" and "Microphones-LO" and set oscillator frequency to 1000 cps. Also make these additional settings on the front panel:

Test Condition switch	to ART REC
Microphone Load	to 1000
Depress Sound Power Telephone Switch	

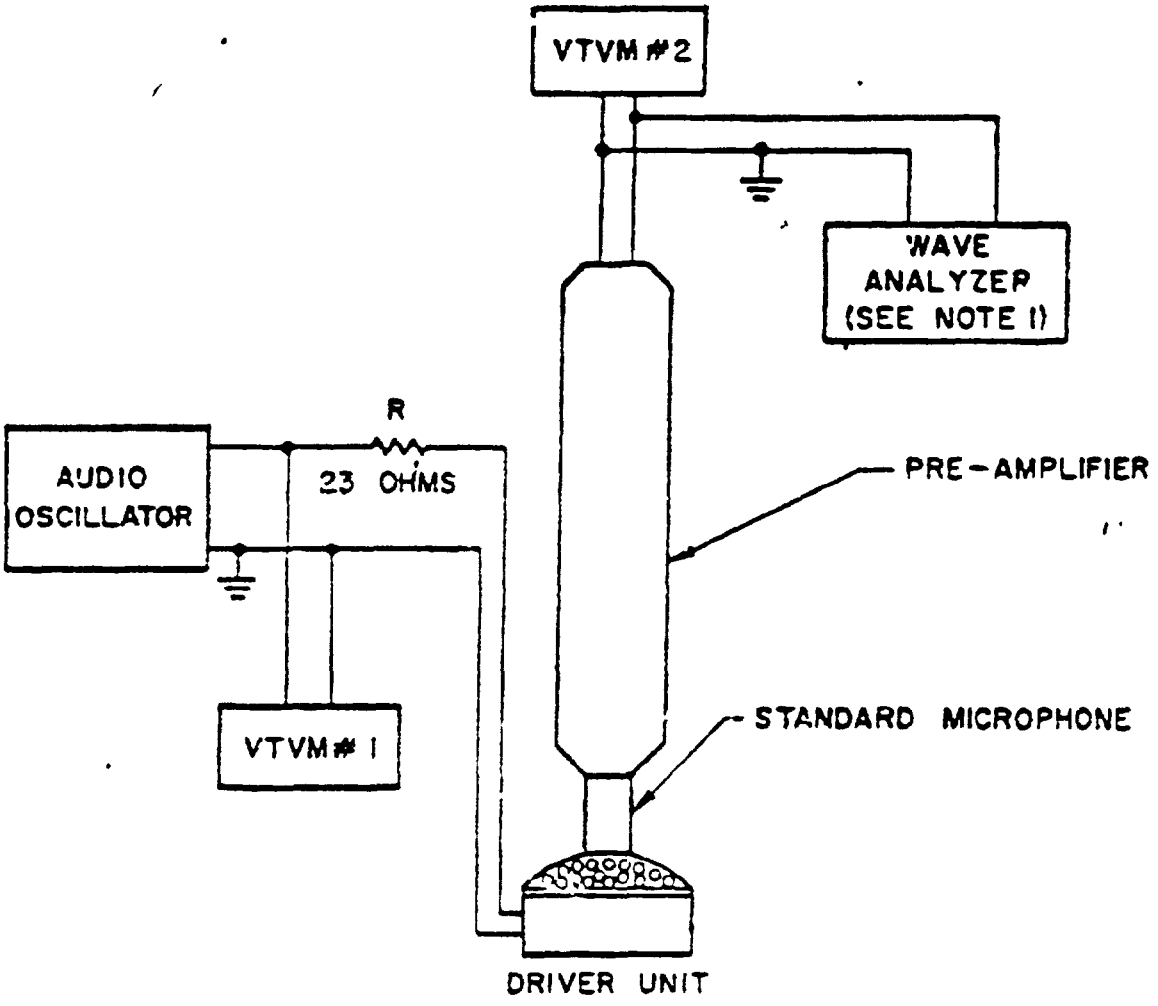
With the VTVM connected to terminals J1L and J1K (ground), adjust oscillator to produce a reading of 300 microvolts at VTVM. This is the input voltage to the pre-amplifier. Now connect the VTVM to terminals J1A and J1K (ground), the VTVM shall show a reading of 60 db ±1 db. This output shall be available over the frequency range of 300 to 5100 cps.

4.10.2 Distortion. - The output of the pre-amplifier while being tested as in 4.10.1 shall be connected to a Hewlett-Packard Model 330-B Total Noise and Distortion Meter, or equal, and distortion measurements shall be made at sufficient points between 300 cps and 5100 cps to determine the frequency where maximum distortion exists. Total harmonic distortion shall meet the requirements of 3.10.2.

4.11 Meter amplifier tests.-

4.11.1 Gain test.- With equipment connected as in 4.10.1, use VTVM to measure the input voltage to the meter amplifier. The output voltage obtained at J2D and J2P (ground) shall indicate a nominal gain of 65 db with both LEVEL - "A" and "B" knobs on front panel set at 0.

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NOTE 1 WAVE ANALYZER REQUIRED FOR DISTORTION TEST ONLY

DRIVER UNIT FREQUENCY RESPONSE TEST CIRCUIT

FIGURE 3

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4.11.2 Meter color band test. - Adjust "Meter Sensitivity" control knob on front panel until needle of meter is at "Calibrate" line. Turning this knob counterclockwise to bring needle to left edge of yellow scale shall give a -6db ±0.5 db reading on VTVM; to left edge of red scale a -3 db ±0.5 db reading, and to left edge of green scale a-1.5 db ± 0.5 db reading. Turning this knob clockwise to right edges of above colors shall indicate meter readings of +6 db ±0.5 db, +3 db ±0.5 db and +1.5 db ±0.5 db respectively.

4.12 Sound source supply test. - Set the front panel controls as follows:

Control	Position
Power Switch	Off
Noise Generator Adjust Switch	Depressed
Noise Generator Calibrate Knob	Fully Counterclockwise
Meter Sensitivity Calibrate Knob	Fully Counterclockwise
Insulation Resistance Knob	Fully Counterclockwise
Dial Speed Knob	Fully Counterclockwise
Dial Break Knob	Fully Counterclockwise
Noise Generator Switch	Driver
Test Condition Switch	Coupler
Microphone Load Switch	Off
Receiver Load Switch	Off
Microphone Current Knob	Off
Level A and B Knob	Any Position
Value C and D Knob	Any Position

Turn on power switch, depress Noise Generator Adjust switch and turn clockwise Noise Generator Calibrate knob until needle of meter is at red CALIBRATE line. With calibrated 640AA microphone connected to vacuum tube voltmeter (VTVM) #2 as in Figure 1, (Driver unit response test) and grid of 640AA microphone lightly touching the screen of the driver unit, the reading of VTVM #2 shall meet the requirements of 3.12.

4.13 Circuit tests.- The front panel controls shall be set in accordance with 4.12 except that LEVEL A knob shall be set at 0, LEVEL B knob at 6, VALUE C knob at 15, VALUE D knob at 21 and NOISE GENERATOR Switch positioned at LOAD. Also, the coupler microphone shall be positioned directly over and touching the screen of the driver unit. Power switch shall now be turned on and the following tests shall be performed in the order indicated below.

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4.13.1 Power-on lamp. - Turn power switch on, and power-on lamp on front panel shall light. (See 3.7.)

4.13.2 Noise generator calibrate knob. - Turn noise generator calibrate knob, needle of meter can be set to red CAL line on meter face. (See 3.12).

4.13.3 Meter adjust switch. - Depress meter adjust switch and turn meter sensitivity knob, needle of meter can be set to red CAL line on meter face. (See 3.11.1).

4.13.4 Hand generator circuit test. - Depress hand generator switch. Meter shall read 0. (See 3.13.)

4.13.5 Ringer circuit test. - Depress ringer switches marked 2400, 5300 and 12000. As each switch is depressed meter shall read 30 or more, and the buzz of the inverter shall be audible. (See 3.14).

4.13.6 Varistor circuit test. - Depress varistor switch. Meter shall read on green scale. (See 3.15).

4.13.7 Earphone circuit test. - Depress earphones switch. Meter shall read on green scale. (See 3.16).

4.13.8 Sound power telephone circuit test. - Depress sound power telephone switch. Meter shall read on green scale. (See 3.16 and 3.17).

4.13.9 Dynamic microphone circuit test. - Depress microphones other types switch. Meter shall read on green scale. (See 3.17).

4.13.10 Ringer circuit test. (repeat). - Depress ringer switches. Meter shall read 30 or more and the buzz of the inverter shall be audible. (See 3.14).

4.13.11 Varistor circuit test. - Depress varistor switch. Meter shall read on green scale. (See 3.15).

4.13.12 Earphone circuit test. (repeat). - Depress earphones switch. Meter shall read on green scale. (See 3.16).

4.13.13 Noise generator circuit test. - Turn noise generator switch to driver. Meter shall read on green scale and noise from driver unit shall be audible. (See 3.8).

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4.13.14 Filter circuit test. - Press Low, Center, High and Narrow Band buttons sequentially. Meter shall read on green scale as each button is depressed. (See 3.18).

4.13.15 Dynamic microphone circuit test (repeat). - Depress microphones other types switch. Meter shall read on green scale. (See 3.17).

4.13.16 Carbon microphone circuit test. - Depress microphones carbon switch. Meter shall read on green scale. (See 3.19).

4.13.17 Dial percent break circuit test. - Depress dial break switch. Needle of meter can be moved to red CAL line when Dial Break Calibrate knob is turned clockwise. (See 3.20).

4.13.18 Dial speed circuit test. - Depress dial speed switch. Needle of meter shall fluctuate slightly and can be moved to red CAL line when Dial Speed Calibrate knob is turned clockwise. (See 3.21).

4.13.19 Insulation resistance circuit test. - Depress insulation resistance switch. Needle of meter can be moved to red CAL line when Insulation Resistance Calibrate knob is turned clockwise. (See 3.22).

4.13.20 Telephone efficiency - Send circuit test. - Depress telephone efficiency-send switch. Meter shall read on green scale. (See 3.24).

4.13.21 Telephone efficiency-Receive circuit test. - Depress telephone efficiency-receive switch. Meter shall read on green scale. (See 3.25).

4.13.22 Continuity circuit test. - Place shorting bar across continuity test leads terminals. Continuity tes. lamp shall light and sound of buzzer shall be audible.(See 3.23).

4.14 Service condition tests. -

4.14.1 Temperature. - The equipment shall be subjected to the temperature cycle shown on Standard MIL-STD-169, amended to conform to the nonoperating limits of 3.26.2. The test of 4.8, 4.9, 4.10, 4.11 and 4.12 shall be performed.

4.14.2 Moisture-resistance test. - The equipment shall be tested as follows:(See 3.26.3).

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4.14.2.1 Test conditions. -

(a) Do not move units from the humidity chamber for measurements.

(b) Start measurements not more than 5 minutes after power is applied to the units. Complete measurements as rapidly as possible. Do not leave power on after measurements have been completed.

(c) Perform the entire test with the chassis removed from its enclosure.

4.14.2.2 Test method. -

(a) Dry at 130° ±5°F for 24 hours.

(b) Condition at 77° ±5°F and 40 to 50 percent relative humidity for 24 hours.

(c) Take measurements as specified in 4.14.2.3 and readjust or realign as necessary to meet full specification requirements.

(d) Subject to continuous cycling for five 48-hour cycles. Temperature, relative humidity and period of time for each portion of the cycle shall conform to Standard MIL-STD-170. The performance measurements shall be made during the periods shown on the standard and shall comply with specification requirements.

(e) After cycling has been completed, condition the equipment for 24 hours at 77 ±5F and 40 to 60 percent relative humidity. Then adjust for optimum performance, using only those means provided by the units. No repair or replacement of parts shall be made. The units shall meet full specified performance.

4.14.2.3 Performance. - Measurements made during test, and specified performance during cycling, shall be in accordance with the tests of 4.8, 4.9, 4.10, 4.11 and 4.12.

4.14.2.4 Failure. - If the equipment fails to meet specified performance during cycling, it shall be readjusted once. If the equipment then fails subsequently during cycle, it does not pass the test. In addition, if the equipment does not meet 4.14.2.3 it does not pass the test.

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4.14.3 Elevation test. - The equipment, placed in its normal operating position in an altitude chamber, shall be tested as follows: (Note: To check temperature stabilization, place a thermocouple on the largest internal mass centrally located within the equipment. (See 3.26.4).

(a) With the equipment placed in the chamber, operating and stabilized at room temperature, perform the tests of 4.8 and 4.9.

(b) Reduce the barometric pressure to 20.6 ±0.1 inches of mercury (simulated 10,000 feet above sea level) and, after temperature and pressure stabilization, repeat the test of 4.8 and 4.9. The equipment shall meet specified performance.

(c) Remove the power from the equipment and further reduce the barometric pressure to 3.4 ±0.1 inches of mercury (simulated 50,000 feet above sea level). Maintain this condition for 1 hour.

(d) Return the chamber to ambient pressure. Apply power to the equipment and, after stabilization, repeat the tests of 4.8 and 4.9. The equipment shall meet specified performance.

4.14.4 Vibration test. - Vibration of the equipment shall be measured as follows, to determine compliance with 3.26.5.

(a) Secure the equipment directly to a vibration table that can be controlled within 10 percent of the pecified amplitude. Mounting method shall be such that vibration within the equipment can be observed and measured. To facilitate this observation and measurement, subassemblies may be tested separately provided they are secured to the table in a manner similar to that used to mount them in the equipment.

(b) Vibrate the equipment successively in three mutually perpendicular directions over a frequency range of 10 to 55 cycles per second. The total excursion of the applied vibration shall be not less than 0.030 inch.

(c) In each of the three directions, change the frequency in steps of one cycle per second and maintain each frequency for at least 10 seconds.

(d) Measure vibration amplitudes by optical means, or by other means provided that vibration of the part is not effected by the measurement.

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4.14.5 Shock test: Bench-handling.- (See 3.26.6) Each assembly which can be serviced on a bench shall be tested as follows: The chassis and front panel assembly shall be removed from its enclosure, as for servicing, and placed in a suitable position for servicing on a solid, 2-inch fir bench top. The test shall be performed as follows, in a manner simulating shocks liable to occur during servicing:

(a) Tilt up the assembly through an angle of 30 degrees, using one edge of the assembly as an axis, and permit the assembly to drop back freely to the horizontal. Repeat, using other practicable edges of the same horizontal face as axes, for a total of four drops.

(b) Repeat "(a)" with the assembly resting on other faces, until it has been dropped for a total of four times on each face on which the assembly might practically be placed during servicing.

4.15 Inspection for dimensional interchangeability.- The dimensions listed below shall be measured to determine conformance to the physical interchangeability requirements of 3.28. When a listed dimension is not within specified or design limits, it shall be considered a major defects.

a. External and internal dimensions of the following:

(1) Case.

(2) Cover.

(3) Printed circuit boards.

(4) Front panel

b. Dimensions of cavities for the following:

(1) Front chassis.

(2) Rear Chassis.

(3) Printed circuit boards.

c. Location of hinges and fasteners on chassis, case and cover.

d. Location of connectors and mounting which receive mating parts of plug-in assemblies assemblies.

4.16 Visual and mechanical inspection.- Equipment shall be examined for the defects listed in Standard MIL-STD-252 and Table V.

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Table V - Classification of Defects

Type defect	Description of Defect
Major	Large burrs, stripped threads and imperfections which will interfere with the assembly for any operation. Missing, inoperative, improperly assembled, defective, or wrong parts which could cause equipment to become inoperative in service.
Minor	Controls do not operate smoothly or freely. Burrs or imperfections on parts which will not interfere with proper operation, assembly or disassembly.

4.17 Quality conformance inspection of preparation for delivery. - Preparation for delivery shall be inspected in accordance with Specification MIL-P-116 to determine conformance to the requirements of Section 5.

5. PREPARATION FOR DELIVERY

5.1 Preservation and Packaging

5.1.1 Level A. - Each Test Set, Telephone TS-716( )/U shall be preserved and packaged in accordance with the procedures specified for the designated methods as prescribed in Specification MIL-P-116 and described in the following paragraphs.

5.1.1.1 Technical Manual. - Each technical manual shall be packaged Method IC-6 as follows: Inclose each technical manual within a close-fitting Type II, Class D bag conforming to Specification MIL-B-117. Heat seal to effect closure.

5.1.1.2 Spare Fuses and Lamps. Package fuses and lamps, Method IIII as follows: (Like stock numbers shall be grouped and packaged together). Cushion each item by wrapping in paperboard, Type I, conforming to Specification PPP-P-291. Secure the wraps with gummed paper tape conforming to Specification UU-T-111.

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5.1.1.3 Stowage within Test Set, Telephone TS-716( )/U.- Stow within the compartments of the Test Set Case, the items packaged as specified in 5.1.1.2 and 5.1.1.3. Close case cover and secure all fastenings.

5.1.1.4 Test Set, Telephone TS-716( )/U. - Each Test Set, Telephone TS-716( )/U shall be packaged Method 1C5 as follows: Cushion each test set on all surfaces with cells or pads or both fabricated of Fiberboard, Type CF, Class weather Resistant, Variety SW, Grade W5c conforming to Specification PPP-F-320. Place the cushioned test set within a close-fitting style RSC, Type 1, Class 2, W5c fiberboard box conforming to Specification PPP-B-636. Place the technical manual packaged as specified in 5.1.1.1 on top of contents directly under lid of the box. Closure shall be made in accordance with the appendix of the box specification. In addition, seal all center seams and end edges and the manufacturer's joint with pressure-sensitive tape conforming to Specification PPP-T-76.

5.1.2 Level C. - Each Test Set, Telephone TS-716( )/U shall be preserved and packaged in accordance with commercial practice and in a manner that will afford protection against corrosion, deterioration and physical damage during direct shipment to the first receiving activity.

5.1.3 Package Performance. - Package testing shall be performed in accordance with the requirements of Specification MIL-P-116. The rough handling and functional tests will be required only when invoked in the bid request or contract.

5.2 Packing.-

5.2.1 Level A. - Place a quantity of Test Set, Telephone TS-716( )/U within a close-fitting nailed wood box, Style 4, Class 2, or Class 2, Type III, Style A, cleated plywood box conforming to Specification PPP-B-621 or PPP-B-601 respectively. Fabricate the box to fit contents snugly. Closure shall be in accordance with the appendix of the box specification.. Gross weight limitation of the shipping container shall not exceed 200 pounds.

5.2.1.1 Metal Strapping. - Shipping containers shall be strapped with flat steel strapping conforming to Specification QQ-S-781, Type 1, Class B, in accordance with the requirements of the appendix of the applicable box specification.

5.2.2 Level B. - Test Set, Telephone TS-716( )/U shall be packed as specified in 5.2.1 except the nailed wood box shall be Class 1 and the cleated plywood box shall be Type IV, Class 1.

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5.2.3 Level C. - Test Set, Telephone TS-716( )/U shall be packed in shipping containers of the type, size and kind commonly used for the purpose, in a manner that will insure acceptance by common carrier and safe delivery at destination. Shipping containers shall comply with the Rules or Regulations of the common carrier applicable to the mode of transportation.

5.3 Marking. - Interior packages and exterior shipping containers shall be marked in accordance with Military Standard MIL-STD-129.

6. NOTES

6.1 Intended use.- Test Set, Telphone TS-716( )/U is intended for use in testing transducers and telephone communication equipment in fourth and fifth echelons.

6.2 Ordering data. - Procurement documents should specify the following:

(a) Title, number and date of this specification and any amendment thereto.

(b) Level of packaging and level of packing required for shipment (Level A, Level B, or Level C).

(c) The specific paragraphs of section 5 which are applicable to the particular procurement.

(d) Preproduction pack(s) as follows:

Makeup of pack(s)
Number of each kind of pack to be submitted
Inspection to be performed thereon.

(e) Marking and shipping of samples.

(f) Place of final Inspection.

(g) Technical literature required. (See 3.30).

(h) Quantity of running spare parts required. (See.3.30) . .).

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6.2 Ordering data.- (Cont'd)

(1) If rough handling and functional test are required.

6.3 Nomenclature.- The parentheses in the nomenclature will be deleted or replaced by a letter identifying the particular design; for example: TS-716W/U. The contractor should apply for nomenclature in accordance with the applicable clause in the contract. (See 1.1)

6.4 Verification inspection.- Verification by the Government will be limited to the amount deemed necessary to determine compliance with the contract and will be limited in severity to the definitive quality assurance provisions established in this specification and the contract. The amount of verification inspection by the Government will be adjusted to make maximum utilization of the contractor's quality control system and the quality history of the product.

6.5 Group C inspection.- Approval to ship may be withheld, at the discretion of the Government, pending the decision from the contracting officer on the adequacy of corrective action. (See 4.5.3.2).

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